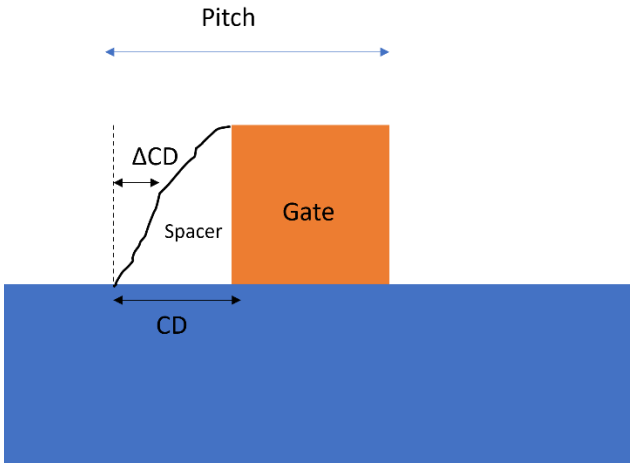
Assignment 5: Spacer



Learning outcomes

1. Importance of spacer in FET fabrication
2. How fins and spacers are oriented

Starting resources

1. Files:
    a. Spacer.zam
    b. Inbuilt example of 14nmFinFET.zam

Questions

At the end of this assignment, we expect a report. Add images or graphs to support your conclusions and answers.

1. Please open Spacer.zam. Build the project. Open the viewer. Understand the structure. Look at the cross-section. Look at the process steps.
    a. Identify the deposition technique used for spacer? Explain why conformality is important here. Name few conformal deposition techniques.
    b. Spacer is patterned without any lithography; give reasons how?
    c. What is the CD loss (ΔCD) for keeping CD =7nm?
    d. Plot CD profile i.e CD variation along gate height
2. Open Inbuilt example of 14nmFinFET.zam.
    a. Can you identify the steps where spacer is deposited and patterned?
    b. What materials are used for spacer?